EC2 Setup
The analysis was performed on an AWS EC2 instance using PySpark. `Java`, `Python`, and required libraries (`pyspark`, `pandas`, `boto3`) were installed. A virtual environment was used to isolate dependencies. Data was accessed from an S3 bucket (`de300-hw3-nyctlc-549787090008-us-east-1-an`) and copied locally for processing.

Dataset & Month of Data
The dataset consists of NYC Taxi and Limousine Commission trip records for *January 2026*, including both yellow and green taxi data. Each dataset contains trip information such as pickup/dropoff timestamps, locations, distance, and fare amounts.

Data Cleaning Assumptions
The following cleaning steps were performed:

1. Removed rows with missing pickup or drop-off timestamps.
2. Removed trips with non-positive distance.
3. Removed trips with negative fares.
4. Removed trips with a negative total amount.
5. Removed trips longer than 24 hours.
6. Removed trips with drop-off time before pickup time.

Analytics Results
<u>Question 1: Which taxi type had more trips?</u>

taxi_type	count
yellow	3518128
green	38909

The yellow taxis had significantly more trips than the green ones.

<u>Question 2: What was the average fare by taxi type?</u>

taxi_type	avg_fare
yellow	21.08260282173478
green	16.003356806908457

The average fare of yellow taxis is $21.08, and the average fare of green taxis is $16.00.

Question 3: What was the average trip distance by taxi type?

```
+---------+-----------------+
|taxi_type|      avg_distance|
+---------+-----------------+
|   yellow|6.757001624157525|
|    green|12.98990593435966|
+---------+-----------------+
```

The average trip distance of yellow taxis is 6.76 miles, and the average trip distance of green taxis is 12.99 miles.

Question 5: What percentage of trips were under 2 miles?

```
Percentage under 2 miles: 51.95206009945919
```

The code calculated that the percentage of trips under 2 miles is 51.95%.

Question 8: Predict the fare amount using non-fare predictor columns.

A Gradient Boosted Tree regression model is used to predict fare amount from a set of feature columns:

- `trip_distance`,
- `pickup_hour`, found by extracting the hour of day from the pickup timestamp,
- `pickup_dayofweek`, found by extracting the day of week from the pickup timestamp,
- `trip_duration`, the difference between the pickup timestamp and the drop-off timestamp,
- `pickup_loc`, the pickup location ID,
- `dropoff_loc`, the drop-off location ID,
- `taxi_type_vec`, the taxi type (`yellow` or `green`) indexed by a `StringIndexer`, then encoded by a `OneHotEncoder`.

80% of the data is used for training, while the other 20% is used for testing. The regressor has a maximum depth of 12 with a seed of 42 for reproducibility. Below are the training and testing RMSEs:

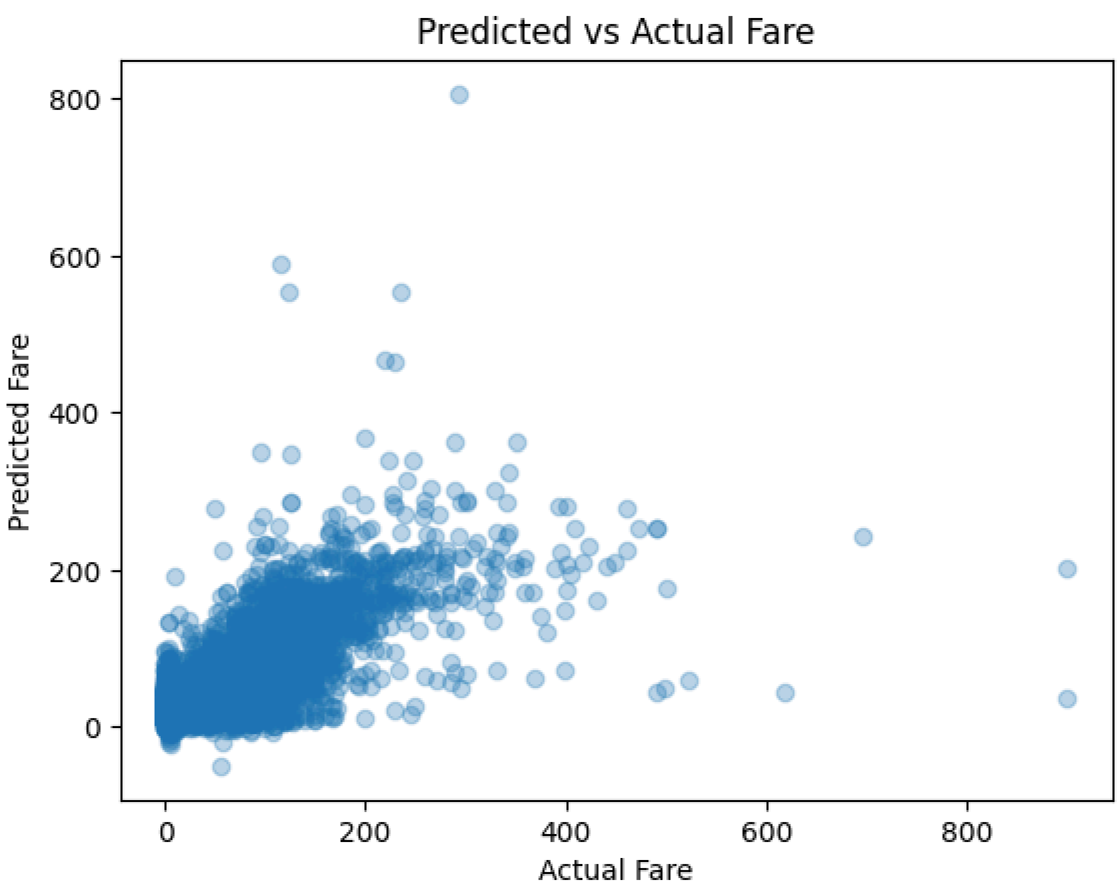
```
Training RMSE: 6.79247972561435
Testing RMSE: 7.60444696873142
```

The model also outputs the feature importances of the predictors:

```
[('trip_distance', np.float64(0.5968941227620662)),
 ('pickup_hour', np.float64(0.05819542504617626)),
 ('pickup_dayofweek', np.float64(0.02903297370578299)),
 ('trip_duration', np.float64(0.16595038504849235)),
 ('pickup_loc', np.float64(0.06964612015340792)),
 ('dropoff_loc', np.float64(0.07638264340895971)),
 ('taxi_type_vec', np.float64(0.003898329875114571 4))]
```

From this list, we can see that the trip distance is the most important factor when predicting the fare amount (having the largest feature importance of 0.5969). Trip duration is also relevant, being the second most important feature. However, the rest are less important. Especially for the taxi type, which has essentially no correlation with the fare amount (having a feature importance of 0.0039).

To further evaluate the performance of this model, a scatter plot is created from the testing predictions vs. actual fare amounts.



Ideally, we would like to see a linear relationship between the actual fare and the predicted fare. The plot shows a somewhat linear relationship, with a few outliers that either have high predicted fare with low actual fare, or high actual fare with low predicted fare. This plot, together with the prediction tables for both the training and the testing datasets, is saved to the S3 bucket `s3://nky8459-de300/nyc-taxi-assignment/`. The structure within this folder is as follows:

- `fare_predictions.png`: This is the scatterplot of predicted vs. actual fare amounts.
- `test_predictions/`: This folder contains the parquet files of the testing dataset predictions.
- `train_predictions/`: This folder contains the parquet files of the training dataset predictions.